# Chapter 1 Preparations

## 1.1 Class, and its Operations

## 1.2 Order Relations

**Definition 1.2.0.1** Consider the following first-order language

- $$\mathcal{L}_{ord,strict} = (M, <)$$

  with: $M$ is a set, $<\subseteq M \times M$, and

  – Irreflexivity

  $$\forall x \neg (x < x)$$

  – Transitivity

  $$\forall x \forall y \forall z ((x < y \wedge y < z) \rightarrow x < z)$$

  Thus,

  $$T_{so} = \left\{ \begin{array}{c} \forall x \neg (x < x) \\ \forall x \forall y \forall z ((x < y \wedge y < z) \rightarrow x < z) \end{array} \right\} \qquad \mathcal{M} \models T_{so} \Leftrightarrow \mathcal{M} \text{ is strictly partial order set}$$

- $$\mathcal{L}_{ord} = (M, \{\leq, =\})$$

  with $M$ is a set, $\leq \ \subseteq M \times M$,

  – Reflecxitive

  $$\forall x (x \leq x)$$

  – Anti-Symmetric

  $$\forall x \forall y (x \leq y \wedge y \leq x \rightarrow x = y)$$

  – Transitivity

  $$\forall x \forall y \forall z ((x \leq y \wedge y \leq z) \rightarrow x \leq z)$$

  Thus,

  $$T_o = \left\{ \begin{array}{c} \forall x \neg (x < x) \\ \forall x \forall y (x \leq y \wedge y \leq x \rightarrow x = y) \\ \forall x \forall y \forall z ((x \leq y \wedge y \leq z) \rightarrow x \leq z) \end{array} \right\} \qquad \mathcal{M} \models T \Leftrightarrow \mathcal{M} \text{ is a partial order}$$

### 1.2.1 Linear-Ordering

**Definition 1.2.1.1** Especially, a partial order $(M, <)$ is called **linear ordered**, if

$$\forall x \forall y (x < y \vee y < x \vee x = y)$$

Thus,

$$\mathcal{T}_{lo} = \left\{ \begin{array}{c} \forall x \neg (x < x) \\ \forall x \forall y \forall z ((x < y \wedge y < z) \rightarrow x < z) \\ \forall x \forall y (x < y \vee y < x \vee x = y) \end{array} \right\} \qquad \mathcal{M} \models T_{lo} \Leftrightarrow \mathcal{M} \text{ is linear ordered}$$

**Well-Ordering**

**Definition 1.2.1.2** Let $(X, \leq)$ be linearly ordered. The relation $\leq$ is called **well-ordering** if every nonempty subset of $X$ has a least element.

$$\forall A \subseteq X, A \neq \emptyset \Rightarrow \exists a \in A \forall b \in A, a \leq b$$

Thus,

$$\mathcal{T}_{wo} = \left( \begin{array}{c} \forall x \neg (x < x) \\ \forall x \forall y (x \leq y \wedge y \leq x \rightarrow x = y) \\ \forall x \forall y \forall z ((x \leq y \wedge y \leq z) \rightarrow x \leq z) \\ \forall A \subseteq X, A \neq \emptyset \Rightarrow \exists a \in A \forall b \in A, a \leq b \end{array} \right) \qquad \mathcal{M} \models \mathcal{T} \Leftrightarrow \mathcal{M} \text{ is well-ordered}$$

## 1.2.2 Well-Founded Order

**Definition 1.2.2.1** The well-founded order is the generation of well-ordering

$$\forall A \subseteq X, A \neq \emptyset \Rightarrow \exists a \in A \forall b \in A, a \leq b$$

## 1.2.3 Density

**Definition 1.2.3.1** ...

# Chapter 2 ZFC Axioms

## 2.1 ZFC Axioms

### 2.1.1 ZFC Axioms

### 2.1.2 Class, Properclass

## 2.2 Axiom of Regularity

♠ **Proposition 2.2.0.1** There exists no set $x \in x$.

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∘ **Proof** ... □

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★ **Corollary 2.2.0.2** There is no pair of sets $x, y$ such that

$$x \in y \qquad y \in x$$

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∘ **Proof** ... □

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### 2.2.1 Von Neumann Hierarchy, Rank

**Definition 2.2.1.1** ...

## 2.3 Axiom of Choice

♠ **Proposition 2.3.0.1** Assume: ZF axiom, the following statements are equivalent:

- The axiom of choice;
- (Choice Function)If $x$ is such that $y \neq \emptyset, \forall y \in x$, then there exists a function

$$f : x \to \cup x \qquad f(y) \in y$$

- (Zorn's Lemma) Let $(x, \leq)$ be a partial order set. Then, suppose
  - $x$ is nonempty;
  - Every chain has a upper bound. i.e.

$$\forall y(y \subseteq x \wedge (a \in y \wedge b \in y \to a \subseteq b \vee b \subseteq a))$$

  Then, $P$ has at least one maximal element $x$ such that: $\forall a(a \in x \to x \not\subset a)$.
- (Hausdorff's Maximal Principle) Let: $X \subseteq P(X)$ be a set, for $Y \subseteq X$, with

$$\forall a \forall b(a \in Y \wedge b \in Y \to a \subseteq b \vee b \subseteq a)$$

  Then, there must some $a \in X$ such that

$$\forall b(b \in Y \to b \subseteq a)$$

  That is, some $a \in X$ such that there is no $b \in X$, with $a \subset b$.

- (well-ordered theorem)Every set can be well-ordered
- (Cartesian Product) The Cartesian product of any family of nonempty sets is nonempty.

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∘ **Proof**

- (2)⇔ (3)
- (2)⇔ (4)
- (2)⇔ (5)
- (2)⇔ (6)

□

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# Chapter 3 Construction of Real Numbers

## 3.1 Natural Numbers

## 3.2 Integer Numbers

## 3.3 Rational Numbers

## 3.4 Real Numbers

† **Axiom** We could construct real number $(\mathbb{R}, +, \times, <) as following$:

- Field Axioms:
- Order Field Axioms:
- Archimede Field:

### 3.4.1 Compeleteness Condition

♠ **Proposition 3.4.1.1** The followings are equivalent:

- Metric space $(\mathbb{R}, |-|)$ is compelete;
- For any upper subset $M \subseteq \mathbb{R}$, $sup(M)$ exists;

#### Cauchy Sequence

◇ **Lemma 3.4.1.2** Suppose $M \subseteq \mathbb{R}$ be upper bounded, then $sup(M)$ exists.

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∘ **Proof** Select upper bound $N$ and $x \in M$,

$$a_0 := x, b_0 = N, c_n := \frac{a_n + b_n}{2} \quad a_{n+1} := \begin{cases} a_n & \forall_{x \in M} c_n \geq x \\ c_n & Elsewise \end{cases} \quad b_{n+1} = \begin{cases} b_n & \exists_{x \in M} c_n < x \\ c_n & Elsewise \end{cases}$$

$(a_n), (b_n)$ are monotone.

Then, $|a_n - b_n| \leq 2^{-(n-1)}|a_1 - b_1| \Rightarrow |b_{n+m} - b_n| \leq |b_n - a_n| \leq 2^{-(n-1)}|b_1 - a_1|$. It is Cauchy sequence. Thus $b_n \to z \in \mathbb{R}$. And $a_n \to z$:

$z$ is $sup(M)$. Suppose $y \in M$, we have

$$y > z \Leftrightarrow \exists N_0 (n > N_0 \to y > a_n)$$

Contradict to the definition. □

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◇ **Lemma 3.4.1.3** If we define $\mathbb{R}$ as

$$\mathbb{R} := 2^{\mathbb{N}}$$

Then,

# Chapter 4 Forcing

Our preparations:

- Models, Languages;
- Filters;

### 4.0.1 Transitive Model

**Definition 4.0.1.1** Recall: Traisitive set;

- A set $M$ is called **transitive** if

$$\forall x \in M, x \subseteq M$$

  equivalently, $\cup M \subseteq M$;

- Consider $\mathcal{L}_{\in} = \{\in\}$, with $\mathcal{M} = (M, E_{\in})$, then, $M$ is called a **traisitive model** of theory $T$ if
  - $M$ is a transitive set;
  - $E = \in|_{M\times M}$;
  - $(M, \in) \vDash T$, where $T$ is a set-theoretic theory;

## 4.1 Forcing Notion

**Definition 4.1.0.1** Take the following notion $\mathbb{P}$ for the **forcing notion**, with

$$\mathbb{P} = (P, \leq, 1_{\mathbb{P}})$$

- $M^{\mathbb{P}}$ by recusion:

$$\begin{aligned} M_0^{\mathbb{P}} &:= \emptyset \\ M_{\alpha+1}^{\mathbb{P}} &= P^M(\mathbb{P} \times M_{\alpha}^{\mathbb{P}}) \\ M_{\alpha}^{\mathbb{P}} &= \cup_{\beta<\alpha} M_{\beta}^{\mathbb{P}} \\ M^{\mathbb{P}} &= \cup_{\alpha\in ord} M_{\alpha}^{\mathbb{P}} \end{aligned}$$

  which is called $\mathbb{P}$**-Names**

- $(P, \leq)$ is a preorder;

Here, $M$ is the universe of set theory.